Electricity monitoring systems

"I'm researching on development of electricity monitoring system in room and appliance level. I want to know the existing research papers and the research gap" *Generated by LeapSpace*

Electricity Monitoring Systems at Room and Appliance Level: Quick Insights

Key Findings Table

Topic	What existing research shows	Most relevant setting	Citations
Core monitoring architectures	Research clusters into intrusive load monitoring (per-appliance sensors), non-intrusive load monitoring (single main meter/NILM), and hybrid systems combining aggregate metering with local sensors or sub-meters [1] [2] [3] [4].	Homes, smart homes, small buildings	[1] [2] [3] [4]
Appliance-level research maturity	Appliance-level monitoring is the most mature stream, dominated by NILM and energy disaggregation methods using HMM/FHMM, CNN, LSTM, TCN, attention, seq2point, and related models [5] [6] [7] [8] [9] [10] [11].	Residential datasets and benchmark studies	[5] [6] [7] [8] [9] [10] [11]
Room-level research maturity	Room-level monitoring exists through circuit breakers, panel/sub-circuit metering, occupancy-linked inference, and spatial context systems, but standardized room-level accuracy reporting is much thinner than appliance-level NILM [12] [13] [14] [15] [16].	Dwellings, institutional buildings, testbeds	[12] [13] [14] [15] [16]

Topic	What existing research shows	Most relevant setting	Citations
Performance trend	Deep learning and multi-modal fusion often improve benchmark accuracy over classical baselines, but gains are highly dependent on dataset, appliance type, and validation setup [11] [17] [18] [19] [20] [21].	Public NILM datasets	[11] [17] [18] [19] [20] [21]
Main unresolved issue	Cross-home, cross-building, and cross-dataset generalization remains a major deployment barrier even when in-dataset accuracy is strong [22] [23] [24] [25] [26] [27] [28].	Real-world deployment	[22] [23] [24] [25] [26] [27] [28]
Practical system trend	Recent systems increasingly use IoT and edge architectures with gateways, MQTT, AMI, Raspberry Pi-class devices, and cloud dashboards for real-time monitoring and control [3] [29] [30] [31] [32] [33].	Smart-home deployment	[3] [29] [30] [31] [32] [33]
Main research gap	The strongest gap is a lack of systems that are simultaneously accurate, low-cost, low-intrusion, privacy-aware, edge-ready, and robust across unseen homes and rooms under standardized evaluation [22] [24] [34] [35] [36].	End-to-end room/appliance monitoring	[22] [24] [34] [35] [36]

Direct Answer

Existing research on room-level and appliance-level electricity monitoring is already substantial, but it is unevenly developed across the two levels. Appliance-level monitoring is the stronger and more mature area, especially through NILM, where studies use single-point metering plus probabilistic or deep-learning disaggregation methods to estimate appliance usage without installing a sensor on each device [1] [5] [8] [11] [37]. Room-level monitoring exists, but it is less standardized and is usually implemented through circuit-level metering, smart breakers, occupancy-linked load analysis, or sensor-fusion systems rather than a unified benchmarked methodology [12] [13] [15] [16] [38] [39].

For developing a new electricity monitoring system, the literature suggests three viable design directions:

1. **Intrusive/per-appliance sensing** for high fidelity but higher cost and maintenance [2] [40].
2. **Pure NILM from a main meter** for low-cost deployment but weaker robustness, especially for overlapping or low-load appliances [1] [39] [41] [42].
3. **Hybrid room/appliance monitoring** combining aggregate power, room/circuit sensing, and contextual sensors, which appears the most promising for balancing practicality and accuracy [4] [38] [39] [43] [44].

The clearest research gap is not “more algorithms” in isolation. It is the absence of a well-evaluated system that integrates **room-level spatial priors**, **appliance-level disaggregation**, **low-cost sensing**, **edge operation**, and **standardized cross-home evaluation** in one deployable framework [22] [34] [35] [36] [45]. Taken together, these findings suggest that a strong new contribution would be a **hybrid, privacy-aware, low-cost room-plus-appliance monitoring system** evaluated on cost, latency, memory, transferability, and privacy exposure, not only accuracy [24] [35] [46] [47] [48].

Confidence level: High - the evidence is consistent across architecture reviews, NILM benchmarks, deployment papers, and evaluation studies [1] [8] [22] [24].

Study Scope

This report covers:

- Foundational NILM assumptions and later monitoring architectures from early single-point disaggregation to current IoT/edge systems [3] [37] [49] [50] [51].
- Intrusive, non-intrusive, hybrid, room/circuit-level, and multimodal monitoring architectures [1] [4] [15] [38] [39].
- Appliance-level disaggregation methods including HMM/FHMM, deep learning, and feature-engineering approaches [5] [6] [8] [9] [52].
- IoT system design, wireless communication, and edge deployment issues [29] [32] [33] [53].
- Datasets, benchmarking, transferability, and open research gaps [22] [24] [36] [54].

Assumptions & Limitations

- The supplied evidence supports a strong thematic review, but not a complete paper-by-paper bibliography with full titles/authors for every citation ID [meta-analisys].
- Room-level monitoring evidence is thinner and less benchmarked than appliance-level NILM, so conclusions about room-level accuracy are more limited [12] [13] [55].
- Privacy and uncertainty are recognized concerns, but explicit evaluation frameworks for them are sparse in the retrieved material [36] [46] [48] [56].

Suggested Further Research

- Build a **hybrid room-aware NILM pipeline** where room/circuit activation constrains appliance candidates during disaggregation [15] [38] [39].

- Develop **standard room-level benchmarks** with clear labels, metrics, and cross-home protocols similar to NILM benchmarking practice [36] [57] [58].
- Study **automatic or low-cost labeling** using appliance-adjacent wireless triggers, occupancy sensors, or contextual metadata [59] [60].
- Compare intrusive, NILM, and hybrid systems on the **same homes** using joint metrics: accuracy, installation effort, cost, maintenance, latency, and privacy risk [24] [35] [40].
- Improve **cross-domain robustness** with transfer learning, domain adaptation, and model compression for edge deployment [26] [27] [28] [33].

Full Report

Introduction and Problem Scope

Electricity monitoring systems differ mainly by **granularity**: whole-home, room/circuit, and appliance level. Whole-home systems measure total demand at one point; room/circuit systems measure demand by zone or branch; appliance-level systems estimate or directly measure individual device consumption [1] [38] [39]. This distinction matters because each granularity supports different applications: billing and overall feedback at whole-home level, occupancy and spatial energy analysis at room level, and device diagnosis, control, and targeted efficiency at appliance level [15] [38] [44].

The modern literature is shaped by a foundational idea from early NILM: appliance usage can be inferred from measurements at a single service-entry point rather than with sensors on every load [37] [49]. Later work retained this single-point framing while extending it with smart-meter data, richer features, open datasets, and reproducible benchmarking tools [36] [50] [61] [62].

Confidence level: High - this framing is repeatedly supported across foundational and review-oriented sources [37] [49] [50].

Foundational Context

Intrusive vs non-intrusive vs hybrid monitoring

- **Intrusive load monitoring (ILM)** attaches sensors or meters to individual appliances, giving more direct measurement but requiring more hardware and installation effort [2] [40].
- **Non-intrusive load monitoring (NILM)** uses one main meter and computationally disaggregates the combined load into appliance-level estimates, reducing intrusion and often cost [1] [63] [64].
- **Hybrid systems** combine aggregate data with sub-meters, smart plugs, room sensors, mobile sensing, or context sensors to compensate for NILM's weak spots [3] [4] [38] [43].

This architecture choice creates a persistent trade-off: more sensing usually improves fidelity, while fewer sensors improve deployability [2] [65] [66].

Room-level vs appliance-level inference

Appliance-level inference aims to identify devices and estimate their energy use. Room-level inference aims to attribute energy to spaces or occupancy zones, sometimes without explicit appliance identification [12] [14] [16]. The literature base is far stronger for appliance-level inference than for room-level inference [13] [67].

Confidence level: High - this asymmetry is directly visible in the evidence mix and benchmark emphasis [8] [58].

Existing Sensing Architectures for Room and Appliance Monitoring

1. Intrusive appliance-level systems

Intrusive systems place sensors on each appliance or load. Reviews describe them as relatively expensive but often more reliable because the measurement is direct rather than inferred [40]. Some comparisons also portray commercial device-based monitoring as practical for consumers, though still reflecting a cost-versus-fidelity trade-off [2]. For research, intrusive sensing remains important because it provides ground truth for training and evaluation [62] [68].

2. Pure NILM systems

NILM remains the dominant research paradigm for appliance-level monitoring because it reduces wiring, sensor count, and installation intrusion [1] [24] [69]. Foundational studies established single-point service-entry monitoring, and later work extended this model to active-power smart-meter data and large benchmark datasets [37] [50] [70]. This stream is algorithmically mature but practically limited by overlapping loads, low-power devices, and domain shift across homes [22] [25] [39].

3. Hybrid room/appliance systems

Hybrid systems are increasingly important. Examples combine whole-home smart-meter data with appliance sensing nodes, local MQTT communication, and cloud storage [3]. Others fuse main-meter readings with mobile sensing or nearby appliance sensors to improve low-load or ambiguous appliance identification [4] [38] [43]. Taken together, these findings suggest that hybridization is the main practical response to the limits of pure NILM [39] [44].

4. Room-level and circuit-level systems

Room-level monitoring is usually implemented through smart breakers, sub-circuit metering, panel-level measurement, or spatially contextual systems [12] [13] [16]. However, many such papers emphasize system design, calibration, dashboards, or real-time summaries rather than comparable room-level accuracy metrics [12] [13]. One notable quantified result comes from occupancy-related load disaggregation in an institutional building, with discrepancy errors reported at 4.84% for lighting, 3.85% for thermal, 1.6% for common loads, and 3.24% for occupancy loads against sub-metered reference data [14].

5. Deployment-oriented hardware

Recent systems increasingly target practical deployment using Raspberry Pi-class gateways, embedded controllers, smart meters, and lightweight edge hardware [22] [30] [51] [71]. This reflects a shift from offline algorithm studies toward operational monitoring pipelines [45] [72].

Confidence level: High - architecture categories and trade-offs are strongly supported across multiple sources [1] [2] [3] [38] [39].

Load Disaggregation Methods for Appliance-Level Inference

Classical and probabilistic models

HMM, FHMM, FHSMM, and related probabilistic sequence models remain important baselines and, in some cases, competitive methods [5] [6] [7]. Reported improvements include lower runtime, better tractability, or stronger accuracy/F1 under specific conditions [6] [73]. Industrial and low-frequency scenarios also show benefits from adding reactive power and duration modeling [74]. However, comparisons across papers are hard because datasets, sampling rates, and metrics differ substantially [18] [75].

Deep learning methods

Deep models such as CNNs, LSTMs, GRUs, TCNs, attention models, seq2point, and seq2seq often outperform classical baselines on public datasets [8] [9] [10] [11] [17] [18]. Representative reported results include strong appliance-specific performance from Seq2Point on UK-DALE, with F1 of 0.92 for a refrigerator and low MAE/RMSE, alongside differing best models for washing machines and air conditioners [19]. CNN-based NILM on REDD has also reported 96.17% mean accuracy for houses 1-4 under its evaluation setting [20]. Self-supervised approaches further aim to reduce dependence on target-house submetering [76].

Feature engineering and signal representation

Despite deep learning progress, feature engineering remains relevant. Harmonics, wavelets, transients, V-I trajectories, and image-based encodings can outperform raw active-power signals when steady-state loads are similar or switching events are discriminative [52] [77] [78] [79] [80]. The trade-off is that these features often require higher sampling rates, more storage, and better metering quality [81] [82].

Benchmarking and generalization

Benchmark papers compare models across datasets such as REDD, UK-DALE, REFIT, Dataport, and AMPds, but they do not identify a single universally best method [83] [84] [85]. Instead, they emphasize appliance-specific difficulty, computational cost, and weak cross-dataset transfer [85] [86] [87].

Confidence level: High - algorithm classes and general benchmark conclusions are consistent across comparative studies [8] [22] [24] [88].

Sensor Fusion, Edge Computing, and System Design

Layered IoT architectures

Residential monitoring systems commonly use layered sensing-to-cloud architectures: sensors or smart meters at the edge, a gateway or microcontroller layer, cloud or server storage, and web/mobile interfaces for monitoring and control [32] [89] [90] [91]. MQTT-mediated communication and dashboard visualization are recurring design patterns [3] [32] [92]. Hybrid smart-home systems may also retain local control when cloud links fail or latency becomes too high [29].

Edge deployment

Edge-native monitoring is emerging as a deployment pattern. A Raspberry Pi edge NILM system reported 17.9 ms inference latency and 312 MB memory usage, supporting the argument that real-time inference is feasible without full cloud dependence [33]. Other embedded or edge systems report low end-to-end latencies and stable long-duration execution [45] [93] [94]. The evidence is stronger for feasibility than for standardized edge-vs-cloud comparative accuracy [33] [94].

Multimodal fusion

Sensor fusion is one of the clearest ways to improve room- or appliance-level attribution. Studies combine meter data with EMF sensors, motion sensors, light and vibration sensors, wireless appliance-adjacent nodes, or BIM context [15] [16] [43] [44] [59]. These additions help with occupancy inference, label generation, and disambiguation of loads that look similar in aggregate power [44] [59]. Inferred: this makes multimodal fusion especially promising for room-aware appliance monitoring, though this integrated formulation is not yet standardized in the retrieved evidence [15] [43].
Confidence level: Low - this is a synthesis across related studies rather than a directly tested unified framework.

Communication and scalability

LoRa, ESP-NOW, AMI, PLC/RF, and similar communication layers are used to improve reach and scalability [53] [95] [96] [97]. The reported trade-off is familiar: LoRa favors long-range and obstacle-rich environments, while ESP-NOW offers better throughput, latency, and energy efficiency for shorter-range deployments [53] [97].

Confidence level: Medium-High - system-design patterns are well supported, but quantitative deployment comparisons are uneven [32] [33] [53].

Datasets, Evaluation Practices, and Research Gaps

Dataset heterogeneity

Datasets vary widely in sampling rate, annotation quality, appliance coverage, synchronization, and ground-truth method [24] [54] [68] [98] [99]. This makes comparison difficult and can strongly influence which methods appear to perform best [23] [66].

Evaluation inconsistency

Benchmarking toolkits and shared protocols such as NILMTK, ECO, and NILMbench were designed to improve reproducibility and comparability [36] [57] [58]. Even so, reviews still report inconsistent metrics, validation settings, and reporting conventions across the field [24] [48] [100].

Generalization failure

A central gap is poor transfer to unseen households, buildings, countries, and appliance mixes [22] [23] [24] [25]. Adaptation strategies under study include meta-learning, transfer learning, domain-adversarial learning, and multi-source domain adaptation [26] [27] [28] [101]. This is one of the clearest signs that benchmark accuracy has outpaced deployment robustness [25] [28].

Cost, latency, and deployment realism

Studies repeatedly show a trade-off among accuracy, computation, memory, sensor cost, and deployment burden [35] [42]. Room-level sensing can become expensive if many meters are needed, while coarse smart-meter data reduces cost but makes disaggregation harder [42] [67]. Some edge frameworks demonstrate that real-time inference is practical if the model is lightweight enough [30] [33].

Privacy

Fine-grained electricity data can reveal occupancy and behavior, motivating encryption, access control, device verification, and local failover [46] [47] [56] [102] [103]. Yet privacy evaluation is underdeveloped relative to accuracy evaluation [48] [104] [105].

Confidence level: High - the gap pattern is repeatedly supported by surveys, benchmarks, and deployment studies [22] [24] [54].

Clear Research Gaps for a New Room- and Appliance-Level Monitoring System

1. Weak integration of room-level and appliance-level inference

The literature has many appliance-level NILM papers and some room/circuit-level systems, but few studies unify them into one scalable framework [8] [12] [13] [67].

2. Lack of robust room-level benchmarks

Room-level monitoring lacks standardized datasets, metrics, and shared evaluation protocols comparable to appliance-level NILM [55] [57] [58].

3. Poor cross-home generalization

Models often degrade in unseen homes and buildings, limiting real deployment value [22] [23] [27] [28].

4. Expensive ground-truth labeling

Appliance-level labels are costly to obtain, especially for supervised training [68] [99]. Hybrid annotation systems using appliance-adjacent wireless triggers are a promising response [59].

5. Limited deployment-oriented comparison

Few studies compare intrusive, NILM, and hybrid systems on the same homes using joint metrics for accuracy, cost, installation time, maintenance, memory, latency, and privacy [35] [40].

6. Underdeveloped privacy-preserving monitoring

Privacy is widely acknowledged, but fewer studies operationalize it as a first-class design or evaluation target [46] [104] [105].

7. Incomplete edge robustness evaluation

Edge-ready systems exist, but long-term fault tolerance, offline resilience, communication failures, and household network constraints remain understudied [29] [33] [94].

Practical Research Direction for Your Project

If your goal is to develop a publishable system, the evidence most strongly supports a **hybrid architecture**:

- **Sensing**: one whole-room or whole-home meter plus low-cost room/circuit sensors and selective appliance-adjacent sensors where ambiguity is high [38] [39] [43].
- **Inference**: room/circuit activation first, then appliance disaggregation within that room as a constrained NILM task [38] [39].
- **System design**: edge gateway with MQTT/local failover plus cloud dashboard for storage and visualization [29] [32] [33].
- **Evaluation**: report not only MAE/F1, but also installation effort, sensor count, memory, runtime, latency, communication reliability, and privacy exposure [24] [35] [48].

Taken together, these findings suggest that the best research opportunity is not a marginally better NILM model on a standard dataset, but a **deployment-realistic, room-aware, appliance-level monitoring system** with standardized cross-home testing [22] [34] [36].

Conclusion

The existing literature is mature for **appliance-level electricity monitoring**, especially through NILM, but still comparatively immature for **true room-level monitoring** [8] [13]. Current work shows clear progress in deep learning, feature fusion, edge deployment, and IoT architectures [11] [33] [45]. However, the major unsolved problem is still the same across streams: achieving strong accuracy **and** practical deployment robustness at low cost and low intrusion [22] [23] [24].

For a new research contribution, the clearest gap is a **standardized, low-cost, privacy-aware hybrid room/appliance monitoring system** that combines spatial priors, NILM, contextual sensing, and edge deployment under realistic evaluation protocols [15] [36] [38] [46] [48].

Overall confidence level: High - the conclusion is supported by converging evidence across architecture reviews, algorithm benchmarks, IoT deployment studies, and evaluation-gap papers [1] [8] [24] [72].

References

1. Moayedi, S., Al Juheshi, F., Almaghrebi, A., (...), Alahmad, M. (2018) An overview of technologies for lower energy consumption in smart buildings. *Proceedings: IECON 2018 - 44th Annual Conference of the IEEE Industrial Electronics Society*. https://doi.org/10.1109/IECON.2018.8592895
2. Benedá, T., Manera, L.T. (2019) Non-intrusive and intrusive energy monitoring methods overview and their relation with household appliances state sensors devices. *Smart Innovation, Systems and Technologies*. https://doi.org/10.1007/978-3-030-16053-1_39
3. Liccardo, A., Bonavolontà, F., Schiano Lo Moriello, R. (2026) A Hybrid IoT Network for Residential Energy Monitoring in Augmented Reality Environments. *Conference Proceedings - 2026 IEEE International Workshop on Metrology for Living Environment, MetroLivEnv 2026*. https://doi.org/10.1109/MetroLivEnv70468.2026.11659869
4. Roux, N., Vrigneau, B., Sentieys, O. (2019) Improving NILM by Combining Sensor Data and Linear Programming. *SAS 2019 - 2019 IEEE Sensors Applications Symposium, Conference Proceedings*. https://doi.org/10.1109/SAS.2019.8706021
5. Salem, H., Sayed-Mouchaweh, M., Tagina, M. (2021) Unsupervised Bayesian Non Parametric approach for Non-Intrusive Load Monitoring based on time of usage. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2020.12.096
6. Wu, Z., Wang, C., Zhang, H., (...), Liu, W. (2021) A time-efficient factorial hidden Semi-Markov model for non-intrusive load monitoring. *Electric Power Systems Research*. https://doi.org/10.1016/j.epsr.2021.107372

7. Rajkumar, N., Govindaram, A., Geetha, R., (...), Anand, A. (2025) Non-Intrusive Load Monitoring Techniques for Intelligent Energy Management: A Comparative Study of FHMM and LSTM Approach. *Proceedings of International Conference on Visual Analytics and Data Visualization, ICVADV 2025*. https://doi.org/10.1109/ICVADV63329.2025.10961750
8. Belguermi, Y., Wira, P., Hermann, G. (2025) Neural Network-Aided NILM (NNAN) disaggregation: Revealing appliance consumption patterns with iterative subtraction. *Machine Learning with Applications*. https://doi.org/10.1016/j.mlwa.2025.100667
9. Akbar, M.K., Amayri, M., Bouguila, N. (2024) Evaluation of Two Novel Supervised Non-Intrusive Load Monitoring Techniques. *2024 IEEE 12th International Conference on Smart Energy Grid Engineering, SEGE 2024*. https://doi.org/10.1109/SEGE62220.2024.10739418
10. Dash, S., Sahoo, N.C. (2022) Multi-Head Attention Based Model for Non-Intrusive Appliance State Detection in Smart Buildings. *1st IEEE Industrial Electronics Society Annual On-Line Conference, ONCON 2022*. https://doi.org/10.1109/ONCON56984.2022.10126909
11. Govula, G.R., Shembe, A.K., Minu, R.I. (2025) A Preliminary Analysis for NILM-Based Energy Management in Residential Microgrids. *Conference Proceedings - 4th IEEE International Conference on Technology, Engineering, Management for Societal Impact using Marketing, Enterpreneurship and Talent, TEMSMET 2025*. https://doi.org/10.1109/TEMSMET65536.2025.11467171
12. Manamperi, K., Shashika, H., Ekanayake, S., (...), Gurusinghe, N. (2021) Smart MCB with metering ability to optimize electricity usage. *2021 IEEE PES/IAS PowerAfrica, PowerAfrica 2021*. https://doi.org/10.1109/PowerAfrica52236.2021.9543109
13. Kiran, S., Khattak, H.A., Butt, H.I., Ahmed, A. (2018) Towards Efficient Energy Monitoring Using IoT. *Proceedings of the 21st International Multi Topic Conference, INMIC 2018*. https://doi.org/10.1109/INMIC.2018.8595549
14. Samadi, M., Fattahi, J. (2021) Energy use intensity disaggregation in institutional buildings – A data analytics approach. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2021.110730
15. Jiang, X., van Ly, M., Taneja, J., (...), Culler, D. (2009) Experiences with a high-fidelity wireless building energy auditing network. *Proceedings of the 7th ACM Conference on Embedded Networked Sensor Systems, SenSys 2009*. https://doi.org/10.1145/1644038.1644050
16. Chiang, C.T., Ho, T.W., Chou, C.C. (2015) A BIM-enabled platform for power consumption data collection and analysis. *Congress on Computing in Civil Engineering, Proceedings*. https://doi.org/10.1061/9780784479247.012
17. Martins, P.B.M., Rodríguez Carneiro Gomes, J.G.R.C., Nascimento, V.B., De Freitas, A.R. (2018) Application of a Deep Learning Generative Model to Load Disaggregation for Industrial Machinery Power Consumption Monitoring. *2018 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids, SmartGridComm 2018*. https://doi.org/10.1109/SmartGridComm.2018.8587415

18. Yan, Y., Zhu, L., Yang, A., (...), Wen, H. (2023) A Review of Industrial Non-Intrusive Load Monitoring. *ICSMD 2023 - International Conference on Sensing, Measurement and Data Analytics in the Era of Artificial Intelligence, Proceedings*. https://doi.org/10.1109/ICSMD60522.2023.10490483

19. Urzagasti, C.A., De Freitas, I.S., Junior, O.H.A., (...), Salvadori, F. (2026) Load Disaggregation in Residential Appliances With Multiple States: Comparative Techniques. *IEEE Access*. https://doi.org/10.1109/ACCESS.2026.3679677

20. Moradzadeh, A., Mohammadi-ivatloo, B., Abapour, M., (...), Rhee, S.-B. (2021) A practical solution based on convolutional neural network for non-intrusive load monitoring. *Journal of Ambient Intelligence and Humanized Computing*. https://doi.org/10.1007/s12652-020-02720-6

21. Ayub, M., El-Alfy, E.-S.M. (2025) Household Appliance Identification Using Vision Transformers and Multimodal Data Fusion. *IEEE Transactions on Consumer Electronics*. https://doi.org/10.1109/TCE.2025.3565850

22. Moghadam, P., Roshan Fekr, A.R. (2026) Non-intrusive load monitoring for smart residential buildings: An AI-based framework with cross-country generalization. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2026.117632

23. Kuloor, A., Singh, A., Dhru, H., Batra, N. (2026) NILMBENCH2026: A Benchmark for Energy Disaggregation. *Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, BuildSys 2026*. https://doi.org/10.1145/3744256.3812587

24. Pereira, L., Nunes, N. (2018) Performance evaluation in non-intrusive load monitoring: Datasets, metrics, and tools—A review. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*. https://doi.org/10.1002/widm.1265

25. Abbas, M.Z., Sajjad, M.I.A., Khan, M.F.N., Martirano, L. (2025) Challenges and Prospects of Different NILM Frameworks for Real-World Applications. *Conference Proceedings - 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe, EEEIC / I and CPS Europe 2025*. https://doi.org/10.1109/EEEIC/ICPSEurope64998.2025.11169169

26. Luo, Q., Yu, T., Lan, C., (...), Pan, Z. (2024) A Generalizable Method for Practical Non-Intrusive Load Monitoring via Metric-Based Meta-Learning. *IEEE Transactions on Smart Grid*. https://doi.org/10.1109/TSG.2023.3263814

27. Pujić, D., Tomašević, N., Batić, M. (2023) A Semi-Supervised Approach for Improving Generalization in Non-Intrusive Load Monitoring. *Sensors*. https://doi.org/10.3390/s23031444

28. Luan, W., Zhang, R., Liu, B., (...), Li, K. (2025) Multisource Domain Adaptation for Nonintrusive Load Monitoring Through Data Distillation. *IEEE Transactions on Instrumentation and Measurement*. https://doi.org/10.1109/TIM.2025.3551567

29. Alhasnawi, Bilal Naji, Jasim, Basil H., Rahman, Zain-Aldeen S. A., Siano, Pierluigi (2021) A Novel Robust Smart Energy Management and Demand Reduction for Smart Homes Based on Internet of Energy. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s21144756

30. Moghadam, P., Roshan Fekr, A.R. (2026) Non-intrusive load monitoring for smart residential buildings: An AI-based framework with cross-country generalization. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2026.117632

31. https://www.sciencedirect.com/science/article/pii/S2045232226051709

32. Sabah, M., Elmitwally, A., Eladl, A.A. (2026) An IoT-Based Real-Time Energy-Management System for Smart Load Control in a Residential Microgrid. *Eng*. https://doi.org/10.3390/eng7080418

33. https://www.sciencedirect.com/science/article/pii/S3004926126003933

34. Liu, Q., Lu, M., Liu, X., Linge, N. (2016) An introduction of non-intrusive load monitoring and its challenges in system framework. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-319-48674-1_23

35. Tang, G., Wu, K., Lei, J., Tang, J. (2016) A simple and robust approach to energy disaggregation in the presence of outliers. *Sustainable Computing: Informatics and Systems*. https://doi.org/10.1016/j.suscom.2016.01.004

36. Batra, N., Kelly, J., Parson, O., (...), Srivastava, M. (2014) NILMTK: An open source toolkit for non-intrusive load monitoring. *e-Energy 2014 - Proceedings of the 5th ACM International Conference on Future Energy Systems*. https://doi.org/10.1145/2602044.2602051

37. Hart, G.W. (1992) Nonintrusive Appliance Load Monitoring. *Proceedings of the IEEE*. https://doi.org/10.1109/5.192069

38. Roy, N., Pathak, N., Misra, A. (2016) Fine-grained appliance usage and energy monitoring through mobile and power-line sensing. *Pervasive and Mobile Computing*. https://doi.org/10.1016/j.pmcj.2016.01.003

39. Roy, N., Pathak, N., Misra, A. (2015) AARPA: Combining Mobile and Power-Line Sensing for Fine-Grained Appliance Usage and Energy Monitoring. *Proceedings - IEEE International Conference on Mobile Data Management*. https://doi.org/10.1109/MDM.2015.64

40. Franco, P., Martínez, J.M., Kim, Y.-C., Ahmed, M.A. (2021) IoT Based Approach for Load Monitoring and Activity Recognition in Smart Homes. *IEEE Access*. https://doi.org/10.1109/ACCESS.2021.3067029

41. Taheri, B., Sedighizadeh, M., Nasiri, M.R., Sheikhi Fini, A.S. (2025) A non-intrusive load monitoring algorithm based on real-time feature extraction and deep learning model. *International Journal of Electrical Power and Energy Systems*. https://doi.org/10.1016/j.ijepes.2025.110637

42. Bongungu, Joseph L, Francisco, Paul W, Gloss, Stacy L, Stillwell, Ashlynn S (2022) Estimating residential hot water consumption from smart electricity meter data. *Environmental Research: Infrastructure and Sustainability*. https://doi.org/10.1088/2634-4505/ac8ba2

43. Rajagopal, N., Giri, S., Bergés, M., Rowe, A. (2013) Demo abstract: A magnetic field-based appliance metering system. *IPSN 2013 - Proceedings of the 12th International Conference on Information Processing in Sensor Networks, Part of CPSWeek 2013*. https://doi.org/10.1145/2461381.2461421

44. Yang, L., Ting, K., Srivastava, M.B. (2014) Inferring occupancy from opportunistically available sensor data. *2014 IEEE International Conference on Pervasive Computing and Communications, PerCom 2014*. https://doi.org/10.1109/PerCom.2014.6813945

45. Alam, T., Rokonuzzaman, M., Sarker, S., (...), Hossain, M.I. (2024) Internet of Things-based Home Automation with Network Mapper and MQTT Protocol. *Computers and Electrical Engineering*. https://doi.org/10.1016/j.compeleceng.2024.109807

46. Birouche, C., Hedir, A., Megherbi, O., (...), Laghrouche, M. (2026) Chaos-Enhanced Cybersecurity for Low-Cost Smart Energy Meters in Smart Grids. *Energies*. https://doi.org/10.3390/en19163810

47. Karst, N., Wicker, S.B. (2016) Secure Key Distribution and Revocation for Advanced Metering Infrastructure. *Distributed Sensor Networks: Second Edition: Sensor Networking and Applications*. https://doi.org/10.1201/b12988-19

48. Rafiq, H., Manandhar, P., Rodriguez-Ubinas, E., (...), Palpanas, T. (2024) A review of current methods and challenges of advanced deep learning-based non-intrusive load monitoring (NILM) in residential context. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2024.113890

49. Farinaccio, L., Zmeureanu, R. (1999) Using a pattern recognition approach to disaggregate the total electricity consumption in a house into the major end-uses. *Energy and Buildings*. https://doi.org/10.1016/S0378-7788(99)00007-9

50. Baranski, M., Voss, J. (2004) Genetic algorithm for pattern detection in NIALM systems. *Conference Proceedings - IEEE International Conference on Systems, Man and Cybernetics*. https://doi.org/10.1109/ICSMC.2004.1400878

51. Shahid, K., Gulzar, M.M., Raza, M.A., (...), Ahmad, H. (2024) Machine Learning based Design of Energy Management System using Non-Intrusive Load Monitoring Strategy. *2024 International Conference on Control, Automation and Diagnosis, ICCAD 2024*. https://doi.org/10.1109/ICCAD60883.2024.10553879

52. Isanbaev, V., Baños, R., Arrabal-Campos, F.M., (...), Alcayde, A. (2022) A comparative study on pretreatment methods and dimensionality reduction techniques for energy data disaggregation in home appliances. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2022.101805

53. Wei, W.J., Sheng Jovian, L.H., Choon How, B.L., (...), Bin Abbas, M.F. (2024) Multi-Protocol Wireless Mesh Sensor Network. *2024 IEEE VTS Asia Pacific Wireless Communications Symposium, APWCS 2024*. https://doi.org/10.1109/APWCS61586.2024.10679304

54. Melo, A., Câmara, M.M., Clavijo, N., Pinto, J.C. (2022) Open benchmarks for assessment of process monitoring and fault diagnosis techniques: A review and critical analysis. *Computers and Chemical Engineering*. https://doi.org/10.1016/j.compchemeng.2022.107964

55. Hernandez, Á., Nieto, R., de Diego-Oton, L., (...), Pérez-Rubio, M.C. (2025) Monitoring Daily Activities in Households by Means of Energy Consumption Measurements from Smart Meters. *Journal of Sensor and Actuator Networks*. https://doi.org/10.3390/jsan14020025

56. Adline Priya, G., Karthick C, K., Dinesh, K., (...), Priyadharshan, S. (2025) IoT-Driven Smart Home Energy Optimization and Safety System. *Proceedings of 8th International Conference on Computing Methodologies and Communication, ICCMC 2025*. https://doi.org/10.1109/ICCMC65190.2025.11140754

57. Hoosh, S.M., Kamyshev, I., Penuela, J., (...), Ouerdane, H. (2026) NILMbench: A novel benchmark for high-frequency NILM regression models. *Energy*. https://doi.org/10.1016/j.energy.2026.141678

58. Beckel, C., Kleiminger, W., Cicchetti, R., (...), Santini, S. (2014) The ECO data set and the performance of non-intrusive load monitoring algorithms. *BuildSys 2014 - Proceedings of the 1st ACM Conference on Embedded Systems for Energy-Efficient Buildings*. https://doi.org/10.1145/2674061.2674064

59. Schoofs, A., Guerrieri, A., Delaney, D.T., (...), Ruzzelli, A.G. (2010) ANNOT: Automated electricity data annotation using wireless sensor networks. *SECON 2010 - 2010 7th Annual IEEE Communications Society Conference on Sensor, Mesh and Ad Hoc Communications and Networks*. https://doi.org/10.1109/SECON.2010.5508248

60. https://www.sciencedirect.com/science/article/pii/S2520894225000826

61. Laughman, C., Lee, K., Cox, R., (...), Armstrong, P. (2003) Power signature analysis. *IEEE Power and Energy Magazine*. https://doi.org/10.1109/MPAE.2003.1192027

62. Kelly, J., Knottenbelt, W. (2015) The UK-DALE dataset, domestic appliance-level electricity demand and whole-house demand from five UK homes. *Scientific Data*. https://doi.org/10.1038/sdata.2015.7

63. Cannas, B., Carcangiu, S., Carta, D., (...), Porcu, P. (2020) Real-time monitoring system of the electricity consumption in a household using NILM techniques. *24th IMEKO TC4 International Symposium and 22nd International Workshop on ADC and DAC Modelling and Testing*. https://www.scopus.com/pages/publications/85096753506

64. Hamid, O., Barbarosou, M., Papageorgas, P., (...), Salame, C.-T. (2017) Automatic recognition of electric loads analyzing the characteristic parameters of the consumed electric power through a Non-Intrusive Monitoring methodology. *Energy Procedia*. https://doi.org/10.1016/j.egypro.2017.07.137

65. Davies, P., Dennis, J., Hansom, J., (...), Ward, L. (2019) Deep Neural Networks for Appliance Transient Classification. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. https://doi.org/10.1109/ICASSP.2019.8682658

66. Rodríguez-Navarro, C., Portillo, F., Martínez-Gil, F., (...), Alcayde, A. (2026) Advances in NILM dataset evaluation: a comparative review within the NILMTK framework. *Electrical Engineering*. https://doi.org/10.1007/s00202-025-03476-y

67. Ploennigs, J., Ahmed, A., Hensel, B., (...), Menzel, K. (2011) Virtual sensors for estimation of energy consumption and thermal comfort in buildings with underfloor heating. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2011.07.004

68. Rodríguez-Navarro, C., Portillo, F., Soler-Ortiz, M., Alcayde, A. (2026) A comparative assessment of open-hardware and commercial energy meters for non-intrusive load monitoring. *Measurement: Journal of the International Measurement Confederation*. https://doi.org/10.1016/j.measurement.2026.122275

69. Pihala, H. (1998) Non-intrusive appliance load monitoring system based on a modern kWh-meter. *VTT Publications*. https://www.scopus.com/pages/publications/0346338443

70. Zoha, A., Gluhak, A., Imran, M.A., Rajasegarar, S. (2012) Non-intrusive Load Monitoring approaches for disaggregated energy sensing: A survey. *Sensors (Switzerland)*. https://doi.org/10.3390/s121216838

71. Ruzzelli, A.G., Nicolas, C., Schoofs, A., O'Hare, G.M.P. (2010) Real-time recognition and profiling of appliances through a single electricity sensor. *SECON 2010 - 2010 7th Annual IEEE Communications Society Conference on Sensor, Mesh and Ad Hoc Communications and Networks*. https://doi.org/10.1109/SECON.2010.5508244

72. Kalezhi, J., Ntalasha, D., Chisanga, T. (2018) Using Internet of Things to Regulate Energy Consumption in a Home Environment. *2018 IEEE PES/IAS PowerAfrica, PowerAfrica 2018*. https://doi.org/10.1109/PowerAfrica.2018.8521060

73. Tokam, L.W., Apeke, S.K., Ouro-Djobo, S.S. (2025) Hybrid HDBSCAN-FHMM Approach for Energy Disaggregation in Non-Intrusive Load Monitoring (NILM) Systems. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3570930

74. Luan, W., Yang, F., Zhao, B., Liu, B. (2022) Industrial load disaggregation based on Hidden Markov Models. *Electric Power Systems Research*. https://doi.org/10.1016/j.epsr.2022.108086

75. Ren, H., Bianchi, F.M., Li, J., (...), Anfinsen, S.N. (2021) Towards Applicability: A Comparative Study on Non-Intrusive Load Monitoring Algorithms*. *Digest of Technical Papers - IEEE International Conference on Consumer Electronics*. https://doi.org/10.1109/ICCE50685.2021.9427652

76. Chen, S., Zhao, B., Zhong, M., (...), Yu, Y. (2023) Nonintrusive Load Monitoring Based on Self-Supervised Learning. *IEEE Transactions on Instrumentation and Measurement*. https://doi.org/10.1109/TIM.2023.3246504

77. Toopchi, Y. (2025) Online nonintrusive load monitoring using radial basis neural network with advanced flexible particle swarm optimization algorithm for residential load. *Franklin Open*. https://doi.org/10.1016/j.fraope.2024.100198

78. Gao, A., Zheng, J., Mei, F., Liu, Y. (2024) Toward intelligent demand-side energy management via substation-level flexible load disaggregation. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2024.123361

79. Lin, Lin, Zhang, Jie, Gao, Xu, (...), Huang, Nantian (2023) Power fingerprint identification based on the improved V-I trajectory with color encoding and transferred CBAM-ResNet. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0281482

80. Nie, Z., Yang, Y., Xu, Q. (2022) An ensemble-policy non-intrusive load monitoring technique based entirely on deep feature-guided attention mechanism. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2022.112356

81. https://www.sciencedirect.com/science/article/pii/S1573048420003556

82. Wang, A.L., Chen, B.X., Wang, C.G., Hua, D.D. (2018) Non-intrusive load monitoring algorithm based on features of V–I trajectory. *Electric Power Systems Research*. https://doi.org/10.1016/j.epsr.2017.12.012

83. Nie, Z., Yang, Y., Xu, Q. (2022) An ensemble-policy non-intrusive load monitoring technique based entirely on deep feature-guided attention mechanism. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2022.112356

84. Francou, J., Calogine, D., Chau, O., (...), Lauret, P. (2023) Expanding variety of non-intrusive load monitoring training data: Introducing and benchmarking a novel data augmentation technique. *Sustainable Energy, Grids and Networks*. https://doi.org/10.1016/j.segan.2023.101142

85. https://www.sciencedirect.com/science/article/pii/S2662998425006749

86. Himeur, Y., Alsalemi, A., Bensaali, F., Amira, A. (2020) Robust event-based non-intrusive appliance recognition using multi-scale wavelet packet tree and ensemble bagging tree. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2020.114877

87. Kaselimi, Maria, Protopapadakis, Eftychios, Voulodimos, Athanasios, (...), Doulamis, Anastasios (2022) Towards Trustworthy Energy Disaggregation: A Review of Challenges, Methods, and Perspectives for Non-Intrusive Load Monitoring. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s22155872

88. Iqbal, H.K., Malik, F.H., Muhammad, A., (...), Chishti, A.R. (2021) A critical review of state-of-the-art non-intrusive load monitoring datasets. *Electric Power Systems Research*. https://doi.org/10.1016/j.epsr.2020.106921

89. Joha, M.I., Nazim, M.S., Zubair, M.I. (2023) Smart Energy Metering, Control, and Protection System for Making a Smart Home Using IoT. *2023 26th International Conference on Computer and Information Technology, ICCIT 2023*. https://doi.org/10.1109/ICCIT60459.2023.10441201

90. Singh, A., Singh, P., Srivastava, N., (...), Gautam, S. (2025) Individual Load Monitoring using Smart Energy Meter: An IoT-based Approach. *Proceedings of 1st IEEE Uttar Pradesh Section Women in Engineering International Conference on Electrical, Electronics and Computer Engineering, UPWIECON 2025*. https://doi.org/10.1109/UPWIECON67212.2025.11390313

91. Nguyen, T.B., Nguyen, T.C. (2023) Design and fabrication of an IoT-based smart electrical meter for residential energy management. *Indonesian Journal of Electrical Engineering and Computer Science*. https://doi.org/10.11591/ijeecs.v30.i3.pp1259-1268

92. Anitha, D., Suchitra, D., Jain, U., (...), Dinda, A. (2026) AI Smart Timer-Based Power Control and Energy Monitoring System for Home Appliances. *SAE Technical Papers*. https://doi.org/10.4271/2026-28-0060

93. Mohamed, N.E.-D.M., El-Dakroury, M.M., Ibrahim, A.A., Nfady, G.A. (2023) IoT Next Generation Smart Grid Meter (NGSM) for On-Edge Household Appliances Detection Based on Deep Learning and Embedded Linux. *5th Novel Intelligent and Leading Emerging Sciences Conference, NILES 2023 - Proceedings*. https://doi.org/10.1109/NILES59815.2023.10296567

94. Ramadan, Ahmed, Shouman, Marwa A., Attiya, Gamal, (...), Ibrahim, Elhossiny (2025) PrivEdge: a hybrid split–federated learning framework for real-time electricity theft detection on edge nodes. *Scientific Reports*. https://doi.org/10.1038/s41598-026-39064-8

95. Esai Vel Murugan, S., Balaji, T.V., Gurubaran, R., (...), Vivin, A.T. (2025) LoRa-Based Smart Energy Metering and Secure Billing System for Rural Areas. *Proceedings - International Research Conference on Smart Computing and Systems Engineering, SCSE 2025*. https://doi.org/10.1109/SCSE65633.2025.11031068

96. BenRhouma, O., Rebai, C., Ben-Romdhane, M., (...), Panzavecchia, N. (2023) The Environmental Impacts of Radio Frequency and Power Line Communication for Advanced Metering Infrastructures in Smart Grids. *Sensors*. https://doi.org/10.3390/s23249621

97. Nicolae, M.-S., Nicolae, P.-M., Nuca, I., (...), Nicolae, R.-C. (2025) Smart Home Monitoring Based on the Lora Protocol as a Solution for Connectivity in Difficult Environments. *SIELMEN 2025 - Proceedings of the 15th International Conference on Electromechanical and Energy Systems*. https://doi.org/10.1109/SIELMEN67352.2025.11260792

98. Klemenjak, C., Reinhardt, A., Pereira, L., (...), Elmenreich, W. (2019) Electricity consumption data sets: Pitfalls and opportunities. *BuildSys 2019 - Proceedings of the 6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*. https://doi.org/10.1145/3360322.3360867

99. Zhang, Y., Qian, W., Ye, Y., (...), Duan, M. (2023) A novel non-intrusive load monitoring method based on ResNet-seq2seq networks for energy disaggregation of distributed energy resources integrated with residential houses. *Applied Energy*. https://doi.org/10.1016/j.apenergy.2023.121703

100. Farah, E., Shahrour, I. (2026) Standardization and Benchmarking of Datasets and Evaluation Protocols for Machine Learning-Based Water Leak Detection: A Systematic Review. *Water (Switzerland)*. https://doi.org/10.3390/w18182267

101. Schirmer, P.A., Mporas, I. (2022) Device and Time Invariant Features for Transferable Non-Intrusive Load Monitoring. *IEEE Open Access Journal of Power and Energy*. https://doi.org/10.1109/OAJPE.2022.3172747

102. Hasan, M.Y., Kadhim, D.J. (2022) Efficient Energy Management for a Proposed Integrated Internet of Things-Electric Smart Meter (2IOT-ESM) System. *Journal of Engineering (Iraq)*. https://doi.org/10.31026/j.eng.2022.01.08

103. Tang, Y., Yi, S., Qin, Z., Li, Q. (2019) Virtual machine migration for IoT applications. *Proceedings of the 4th ACM/IEEE Symposium on Edge Computing, SEC 2019*. https://doi.org/10.1145/3318216.3363383

104. Pereira, L., Nunes, N. (2020) Understanding the practical issues of deploying energy monitoring and eco-feedback technology in the wild: Lesson learned from three long-term deployments. *Energy Reports*. https://doi.org/10.1016/j.egyr.2019.11.025

105. Kahl, M., Krause, V., Hackenberg, R., (...), Udalzow, A. (2019) Measurement system and dataset for in-depth analysis of appliance energy consumption in industrial environment. *Technisches Messen*. https://doi.org/10.1515/teme-2018-0038